

Basic Principles of Medicine 1

Module : Musculoskeletal System

MSK DLA

DLA Title: Triplet Repeat Disorders & Pharmacogenetics: Pharmacogenetics

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Learning Objectives:

SOM.MKIII.BPM1.2.MSK.1.GNET.0174	Outline the classic examples of gene-drug interactions
SOM.MKIII.BPM1.2.MSK.1.GNET.0175	Compare and contrast the terms pharmacogenetics and pharmacogenomics
SOM.MKIII.BPM1.2.MSK.1.GNET.0176	Explain the pathophysiology of malignant hyperthermia
SOM.MKIII.BPM1.2.MSK.1.GNET.0177	Analyze the genetic aspects of malignant hyperthermia including autosomal dominant inheritance with reduced penetrance as well as locus vs. allelic heterogeneity
SOM.MKIII.BPM1.2.MSK.1.GNET.0178	Describe the cytochrome P450 family of enzymes and their general function
SOM.MKIII.BPM1.2.MSK.1.GNET.0179	Discuss the CYP2D6 enzyme and how it metabolizes metoprolol and codeine

Important Definitions/Terms:

- **Pharmacokinetics** – How a drug is absorbed, distributed, metabolized and excreted.
- **Pharmacodynamics** – The pharmacological effect and clinical outcome.
- **Pharmacogenetics** – Describes a variation in a single gene and how it influences a response to a single drug.
 - The examination of the genetic basis for the individual variation in drug responses.
- **Pharmacogenomics** – A broad term taking into account how all of the genes in a genome influence responses to a drug.
 - Useful in the development of new drugs and identification of new targets.
- Foundation of **Personalized/Individualized Medicine**.

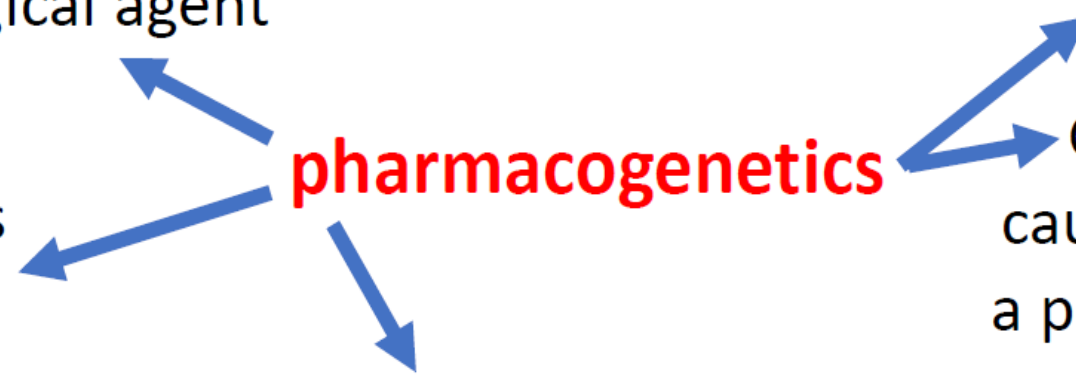
Gene-Drug Interactions:

Malignant hyperthermia:
Phenotype is seen only if the
patient is exposed to a particular
pharmacological agent

Warfarin dosing is
affected by two
genes: one gene
encodes the target
of the drug
(*VKORC*), and
another is involved
in its excretion
(*CYP2C9*)

Codeine sensitivity: effect
of a standard dose varies
depending on alleles of
CYP2D6 which encodes a
p450 enzyme that
converts codeine to its
active form (morphine) .

pharmacogenetics

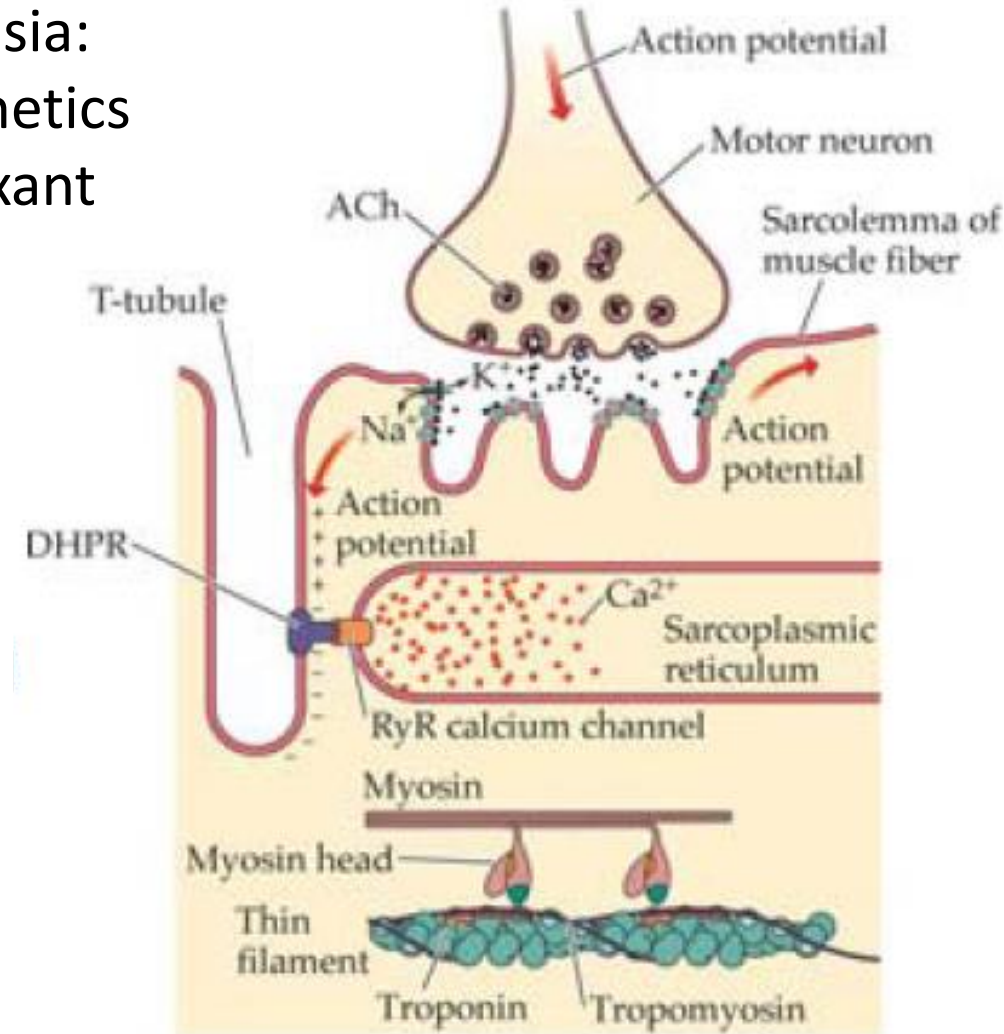


6-mercaptopurine dosing:
dosage will vary depending
on genotype of patient

G6PD deficiency: may
cause hemolytic anemia in
a person who is exposed to
oxidizing agents - an
idiosyncratic effect.

Malignant Hyperthermia:

- Associated with an adverse reaction to general anesthesia:
 - ❖ Halothane – volatile halogenated inhalation anesthetics
 - ❖ Succinylcholine – depolarizing skeletal muscle relaxant
- 80% associated with mutations in Ryanodine receptor gene (*RyR1*) on Chr 19 – plant toxin ryanodine.
- Together with dihydropyridine receptor – function in controlled release of Ca^{2+} from sarcoplasmic reticulum.
- Mutation in RyR – excessive release of Ca^{2+} and sustained muscle contraction ($\uparrow$ membrane depolarization, $\uparrow$ respiration).
- Clinical symptoms - muscle rigidity (CPK & myoglobinuria), hypertension, hyperthermia, acid/base imbalance leading to acidosis, hyperkalemia and tachycardia.

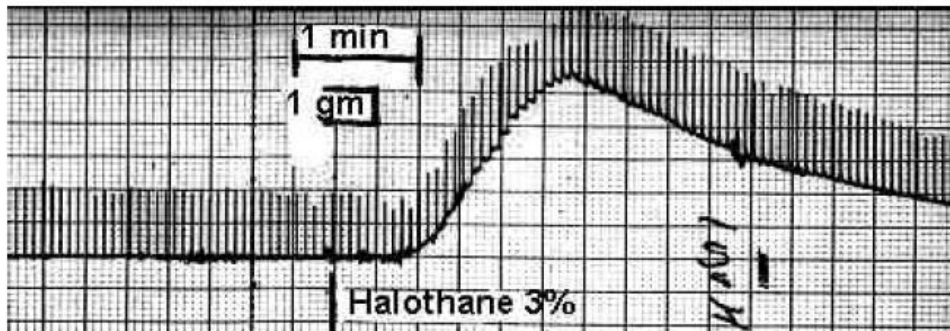


Detection of MH:

- Knowledge of the gene not necessary – RyR >100 exon (5000aa), multiple mutations.
- Most reliable test - caffeine-halothane muscle contracture test (small muscle biopsy).



- Abnormal muscle produces a response of > 0.2g to 2mM caffeine.
- Response of 0.6g to 1mM caffeine, 1.6g to 2mM.



Abnormal halothane test

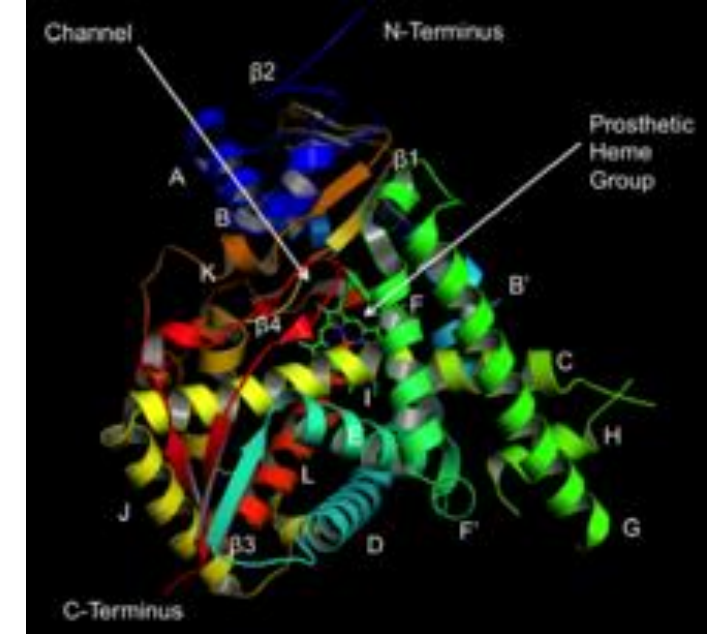
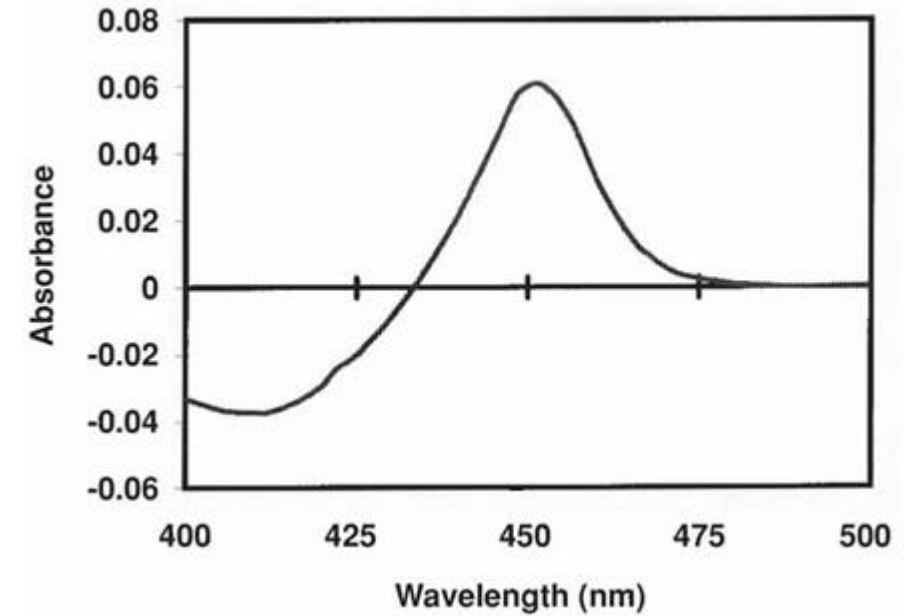
- For halothane response use 3% dose.
- Normal muscle will not change baseline by more than 0.5g (1/2 box).

Genetic importance of MH:

- Malignant Hyperthermia – **autosomal dominant trait**.
- Rare with incidence of approx. 1:15,000
- Most cases associated with mutations in *RyR1* (80%), other with Ca^{2+} channel (1%) and 19% yet to be identified = **locus heterogeneity**.
- *RyR1* – large gene containing multiple mutation types in different individuals = **allele heterogeneity**.
- Pedigree studies reveal evidence of **reduced penetrance** – family history = increased risk!
- Above traits mean screening is not an option – false negatives cannot be tolerated for potentially lethal condition!!!

Cytochrome P450s (CYPs):

- Cytochrome P450s – large family of heme containing enzymes, approx. 50kDa, named due to characteristic CO-reduced absorption spectrum.
- Functions:
 - Drug metabolism
 - Cholesterol biosynthesis
 - Steroid hormone biosynthesis
 - Lipid peroxidation
- CYPs required molecular oxygen and NADPH as a source of electrons – function as a complex with POR (P450 oxidoreductase).
- Perform monooxygenase reaction – introduce oxygen into molecules = drug more water soluble, less toxic and easier to excrete in urine



https://en.wikipedia.org/wiki/Cytochrome_P450

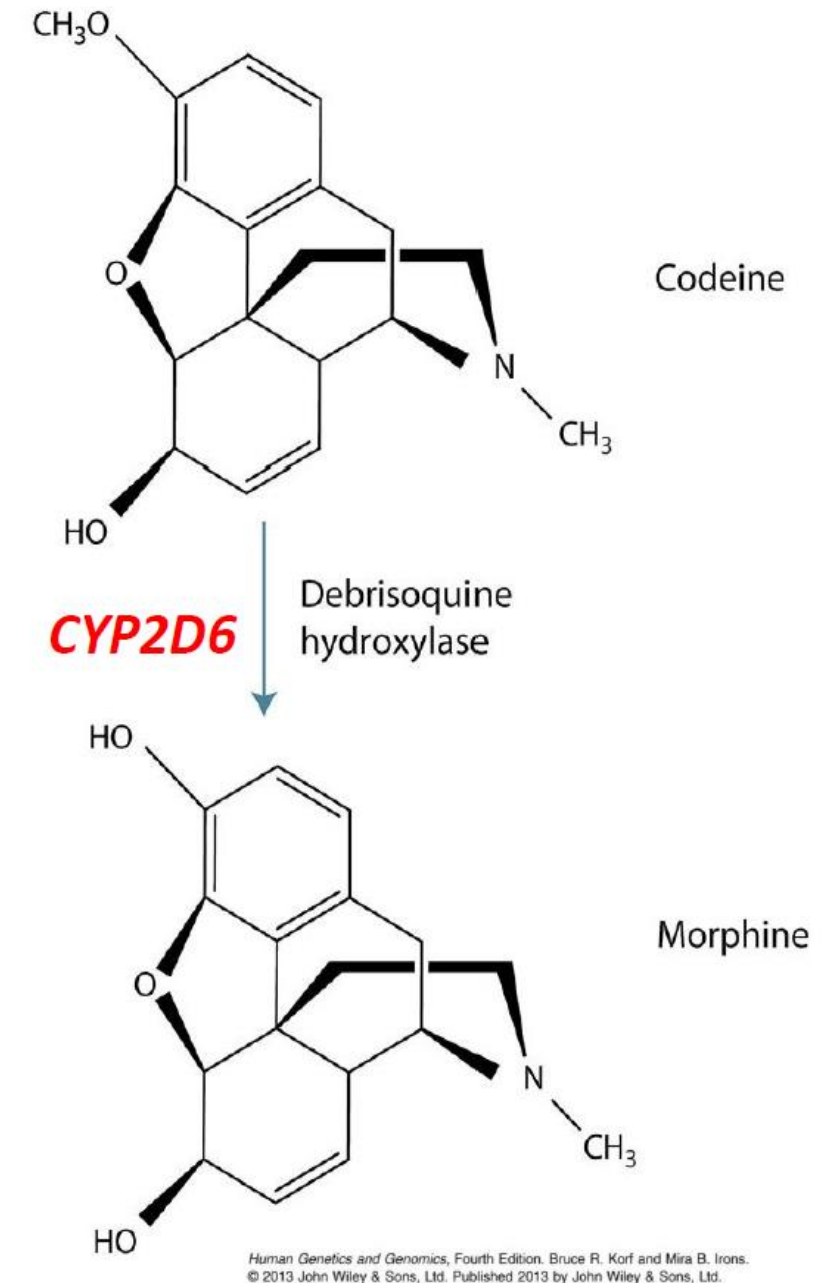
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CYP2D6 and Metoprol:

- CYP2D6 (debrisoquine hydroxylase) - metabolizes a large range of drugs e.g. antidepressants (fluoxetine), antiarrhythmics (propafenone), analgesics (codeine), Beta-blockers (metoprolol).
- CYP2D6 is highly polymorphic – individuals divide into 3 groups:
 - ❖ Poor metabolizers = 1-5% of the population, homozygous recessive for low activity
 - ❖ Extensive metabolizers = 90% of the population, homo/heterozygous for normal activity
 - ❖ Ultra-rapid metabolizers = 5-10% of the population, gene amplification
- CYP2D6 detoxifies metoprolol by addition of a hydroxyl group:
 - ❖ Poor metabolizers = slow metabolism, maintain high level, adverse reaction
 - ❖ Extensive metabolizers = normal metabolism, normal effect
 - ❖ Ultra-rapid metabolizers = rapid breakdown, no effect, require higher dose!

CYP2D6 and Codeine:

- CYP2D6 converts codeine into Morphine – biotransformation.
- Pain relief effect of codeine dependant on this reaction.
- CYP2D6 polymorphism/pharmacogenetic effects:
 - ❖ Poor metabolizers = slow metabolism, ineffective conversion to morphine – no relief
 - ❖ Extensive metabolizers = normal metabolism, normal effect
 - ❖ Ultra-rapid metabolizers = rapid metabolism, highly efficient conversion to morphine – overdose situation = respiratory depression/arrest even from standard dose



Thank you for listening!